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Labour market policies in the age of AI

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1. Embracing AI without fearing “the end of work”

While AI raises important concerns, for instance regarding its ability to influence personal or political decisions and the power it gives to certain firms or individuals, we believe, based on our own research and reading of existing evidence, that the belief that it will necessarily lead to massive unemployment is at least questionable.

The debate is still active. The more pessimistic studies usually reason at the task level in a static way. That is: they consider existing occupations, break them down into tasks and compute the share of tasks that AI can perform. Inevitably, this results in very high numbers.

However, there are reasons why we view this approach as flawed. First, the task content of a given occupation is not fixed over time. The technology will change what we do concretely when we work: as some tasks are taken over by AI, individuals will reallocate their time and effort towards other tasks (Humlum & Vestergaard, 2026). Doctors will use the time saved by the technology to see more patients, white-collar workers will spend less time preparing presentations and more time finding new clients, etc.

Second, jobs are bundles of tasks and some cannot be easily unbundled and require a human to perform the residual tasks (Garicano et al., 2026). Radiology can illustrate this point: a decade after it was announced that AI would make radiologists obsolete, they have never been so numerous, because we still need a human to arbitrate difficult cases, advise surgeons and sign off on a diagnosis.

Third, the substitution effect of human work is only one side of the story. The other is productivity. Of course, large productivity gains on individual tasks need not translate one-for-one into greater output: other tasks or organizational constraints can become bottlenecks, as emphasized by Jones and Tonetti (2026) and illustrated empirically by Demirer et al. (2026). But insofar as AI does raise productivity at the firm level, firms will generally not use those gains simply to produce the same output with fewer workers. They can become more competitive, gain market share

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and expand production, which leads them to hire more, not less. In other words, for AI, as in previous technological revolutions, there are two forces playing in opposite directions: on the one hand the technology displaces human work but on the other hand it allows growth, which pushes employment upwards (Aghion et al., 2026). The key question is the net effect and history tells us there is no reason to be very pessimistic. As the technology improves, the substitution effect increases, but so does the productivity effect. Conceptually, there is no reason to believe that in the steady state the net effect should be negative.

Last but not least, the AI revolution makes new ideas easier to find, and new ideas often translate into new job creations.

Of course, this does not mean that there are no concerns regarding the impact of AI on labor markets. First, there can be a timing issue. The displacement force tends to manifest itself before the positive effects and the creation of new occupations or jobs enter in full force. This can create a temporary drop in employment, with potentially long-lasting consequences for those affected.

Second, there are distributional issues. The fact that overall technology does not reduce labor demand does not mean that it will be beneficial for everyone: there will be winners and losers. These distributional effects may also show up in wages rather than employment. Even if AI has little effect on aggregate employment, it could substantially change the relative wages of workers depending on whether their skills are complementary to or substitutable with the technology.

In the past, high-skilled workers have benefited from technological change because their work was complementary with the new tools while manual or clerical workers performing routine work were more subject to the substitution force as their work tends to be more substitutable with capital. With AI, the jury is still out in terms of which part of the distribution will witness employment growth and/or relative wage premiums and which part will be associated with relative employment declines and/or downwards wage pressure. For the first time, occupations from the top of the distribution are affected by the substitution force. However, we believe this is good news as such workers are more likely to be able to adjust their work to keep their job relevant or to rebound if their occupation is fully displaced.

There might also be some heterogeneity along the age dimension. New studies tend to show negative effects for younger workers. Evidence from the US suggests that, as of today, while economy-wide jobs remain stable, early-career workers (ages 22–25) face steep employment drops in

occupations heavily exposed to generative AI (Brynjolfsson et al., 2026). Ongoing work suggests that these findings replicate in some European countries as well. In France, Palladino et al. (2026) document an aggregate decline in early-career hires in exposed occupations, and in Denmark, Humlum and Vestergaard (2026) find similar declines across several exposed occupations. However, in Norway, Hernæs and Kostøl (2026) find that employment has grown more slowly in exposed occupations than in less exposed ones but with no specific trend for early-career jobs. In any case, we remain cautious in our interpretation of all this evidence. Indeed, the effects are observed from late 2022 onward, at the time of ChatGPT's release, which also coincided with the exit from some Covid era policies, as well as the rise in central bank interest rates. Moreover, hiring in new occupations will likely take more time to materialize.

But even if we can't be sure yet of the distributional effects, we know there will be some. And they should be taken very seriously and call for appropriate policies, as explained in the next section. The last three decades have precisely failed to make sure globalization and the previous technological revolution were working for all workers: this has led to many citizens' discontent with their labor market situation reflected in extreme electoral outcomes.

An additional point of attention pertains to the quality of work: AI agents might be more efficient and more docile to manage than humans but interacting mostly with agents may create a feeling of isolation at work, potentially lowering employees' job satisfaction and well-being, or even their mental health. Similarly, having too many tasks delegated to the technology can lead to a loss of sense of purpose.

However, these concerns should not obscure the fact that there is no reason to believe that in the long run AI will reduce the number of work opportunities for humans. The fear of AI displacing human work is misplaced: it is not grounded in any convincing conceptual or empirical evidence, and it is costly, since it can slow down adoption at a time when Europe precisely suffers from adopting new technologies too slowly.

2. Appropriate policies for labor adjustment

2.1 At the national level : flexicurity and good education

As mentioned, despite our cautiously optimistic view of the effect of AI on employment at a macroeconomic level, we recognize that some people will mainly face headwinds due to AI. Even under fiscal pressure, governments should ensure that they maintain an appropriate level of insurance for workers facing displacement. Indeed, job loss can be a devastating event with long-lasting consequences, including spillovers onto other household members and negative externalities for society. But this is not a law of nature: it depends on the context and in particular on institutions and policies. Our research has shown that job loss can be made a non-event, as was the case in Denmark with its flexicurity model thanks to which none of the negative effects of displacement on health and mortality that are documented in the US take place (Roulet, 2020). It is very important to make sure that job loss is appropriately insured, to make sure the economy can embrace safely technological progress and the job transitions that come with it, while avoiding for instance the “deaths of despair” (Case & Deaton, 2020) and other social ills that can come with prolonged periods of reduced work opportunities.

Flexicurity means income insurance combined with active labor market policies, such as training, monitoring and counseling. These active policies have been extensively studied by labor economists, and the findings point to highly heterogeneous effects (Card et al., 2018) with average effects not necessarily passing a standard cost-benefit analysis. It is therefore important that the resources of public employment services be targeted appropriately towards those for whom benefits are meaningful.

When job seekers register to claim unemployment insurance, standard administrative data combined with appropriate survey questions now make it possible to predict who is at risk of long-term unemployment (Mueller & Spinnewijn, 2025). Presumably, these are the individuals who would benefit most from what public employment services can offer: intensive counseling, training programs, and so on. The allocation of job seekers to such services should be evaluated and improved continuously.

However, we believe traditional flexicurity policies might not be enough. Two other pillars seem to us essential. First, education. We know that job transitions will be more frequent. It is very hard to predict which skills will be most rewarded in the future and this uncertainty can be a hindrance to skill acquisition. Yet the ability to adapt and learn new skills will be key to labor-market success, and good education teaches you how to learn, i.e. makes you more adaptable. It is therefore very important to make sure all citizens have access to an initial education of appropriate quality.

In this respect, the effect of AI on education deserves close attention. While AI offers tremendous opportunities for both students and teachers,

to improve their understanding and their teaching materials respectively, its use in the classroom should be approached with caution. Focusing on Chinese students in grades 7-12, Stromberg et al. (2026) show that, while AI adoption improved homework quality, it lowered exam scores within a few months, with the largest losses in social sciences, STEM, and languages. Thus a whole part of schooling needs to take place without AI: reading books, writing, grammar, calculus, mathematical proofs - all those activities should be performed first without resorting to AI.

Moreover, one should rehabilitate manual jobs – plumber, cook, etc - and develop higher education programs which put more emphasis on manual and technical skills but also include bridges between those and more general higher education curricula following the Swiss model.

The second essential pillar, beyond flexicurity, is industrial policy to promote “good jobs” (Rodrik and Sabel, 2019). And this is where Europe comes in. While flexicurity and education are typically policies decided within member states, Europe is the relevant level for labor-enhancing industrial policies. The direction of innovation responds to expected market size, and no member state offers a market large enough to make a labour-augmenting mandate worth reorienting an industrial policy around. Innovation is also path-dependent, so what Europe is capable of building in 2050 will be largely settled by the incentives it sets this decade.

2.2 At the European level: labor-enhancing industrial policies

Let’s explain what we mean by “labor-enhancing industrial policies”. Even though we should not fear the effects of AI on employment, Europe should still have a strategic vision of the kinds of jobs it wants. Technology is a means, not an end. Market forces will lead to the fastest developments in areas where profit expectations are highest. These may or may not coincide with optimal uses of the technology from society’s point of view. For instance, AI is a blessing for medical progress and waste management, but it may create addictions hampering children’s learning or threatening teenagers’ mental health. Similarly, different uses of AI may either enhance or undermine meaningful and rewarding employment.

In cases where private incentives would not bring about the social optimum, as with carbon emissions (Aghion et al. 2016), it is the role of the government to intervene to better align the two. The same logic of externality can apply for the labor market effects of AI: industrial policies can incentivize the development of AI uses that are labor-enhancing, rather than labor-replacing because of the all the associated social consequences.

The instruments at our disposal are well known, and we can borrow them from climate policy. They include subsidies and tax credits that are conditional on the type of usage. They also include public procurement, for example in health, education, and public administration, where the need for improved services is growing and, in the case of health and education, where labor is already hard to find, which limits the risk of job displacement and can improve working conditions. They finally include the public provision of complementary assets, data, and training, which are necessary for a successful adoption of technology, that is, an adoption that increases productivity and augments labor rather than substitutes for it with lower performance. Such policies should, however, be conducted carefully and avoid having government pick winning technologies and producers, but instead define the right set of institutions to change the direction of technology and let the market and competition operate a selection.

For a long time, European countries favored R&D policies, in which the government chose only the type of expenses eligible for tax support, leaving firms fully in charge of their technological choices. While only 22% of European countries had an R&D tax credit in 2000, 72% had one by 2019. Yet these policies showed their limits: domestic expenditure on R&D rose only weakly within the EU, and the continent missed the digital turn, with the sector's leading firms being American or Chinese. The 2010s marked a shift, with the emergence of mission-oriented industrial policies in Europe, in which the public authorities define the mission and competitive calls for projects select the winners. Ongoing research (Aghion et al., 2026) shows that competitive calls for innovation projects can even be self-financing in the short term. This kind of industrial policy is exactly what Europe needs to be stronger in 2050 and the design of the missions must include a reflection on the labor market impact.

Conclusion

As technology advances, gains in job quality and wage progression are possible, including for low-skilled workers (Aghion et al., 2024; Autor, 2024). But this depends on the economy developing and adopting technology in the right direction: not by impeding technological progress, but by ensuring that market incentives do not steer it toward uses that substitute for labor without raising productivity. At the national level, member states should provide an appropriate level of insurance for workers facing displacement and education policies to develop the ability

to adapt and learn new skills. But this will not be enough: Europe must have a strategic vision of the jobs it wants in 2050 and use mission-oriented industrial-policy instruments to promote these jobs. This can only be done effectively at the European level because market size is key to effectively re-orient the direction of innovation or growth.



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Philippe Aghion is a Professor at the College de France and at the London School of Economics, and a fellow of the Econometric Society and of the American Academy of Arts and Sciences. His research focuses on the economics of growth. With Peter Howitt, he pioneered the so-called Schumpeterian Growth paradigm which was subsequently used to analyze the design of growth policies and the role of the state in the growth process. Much of this work is summarized in their joint books Endogenous Growth Theory (MIT Press, 1998) and The Economics of Growth (MIT Press, 2009), in his book with Rachel Griffith on Competition and Growth (MIT Press, 2006), and in his survey "What Do We Learn from Schumpeterian Growth Theory" (joint with U. Akcigit and P. Howitt.) In 2001, Philippe Aghion received the Yrjo Jahnsson Award of the best European economist under age 45, and in 2009 he received the John Von Neumann Award.

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